Making Qualitative Data Reusable

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27th November 2024

Presentation available under CC-BY Licence at https://doi.org/10.5281/zenodo.14140251

About DANS

DANS

Dutch national centre of expertise & repository for research data

Offers **data publishing and archiving** services for individual researchers and institutes

Offers **advice** and **training** on Research Data Management (**RDM**), **FAIR data** and **Open Science**

Part of CESSDA (Consortium of European Social Science Data Archives)

CESSDA provides large-scale, integrated and sustainable **data services** to the **social sciences** community:

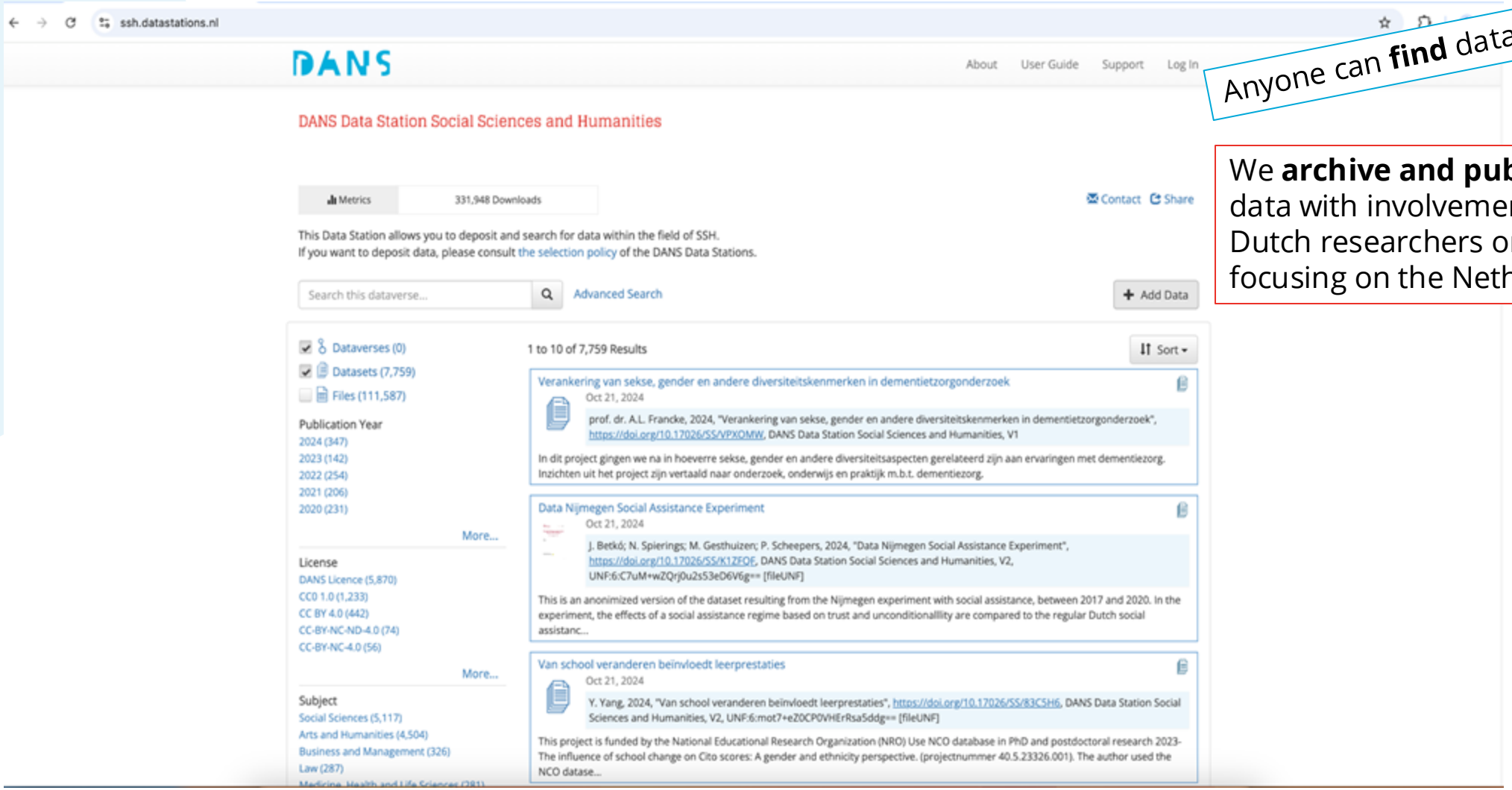
- A Data Discovery Portal — cessda DC Data Catalogue
- Data Management Training — cessda DMEG Data Management Expert Guide
- Controlled Vocabulary Services — cessda ELSST Thesaurus

https://www.cessda.eu/

- **DANS** is CESSDA Service Provider for the Netherlands

  LiDA (Lithuanian Data Archive for Humanities and Social Sciences) for Lithuania

DANS Data Station Social Sciences and Humanities

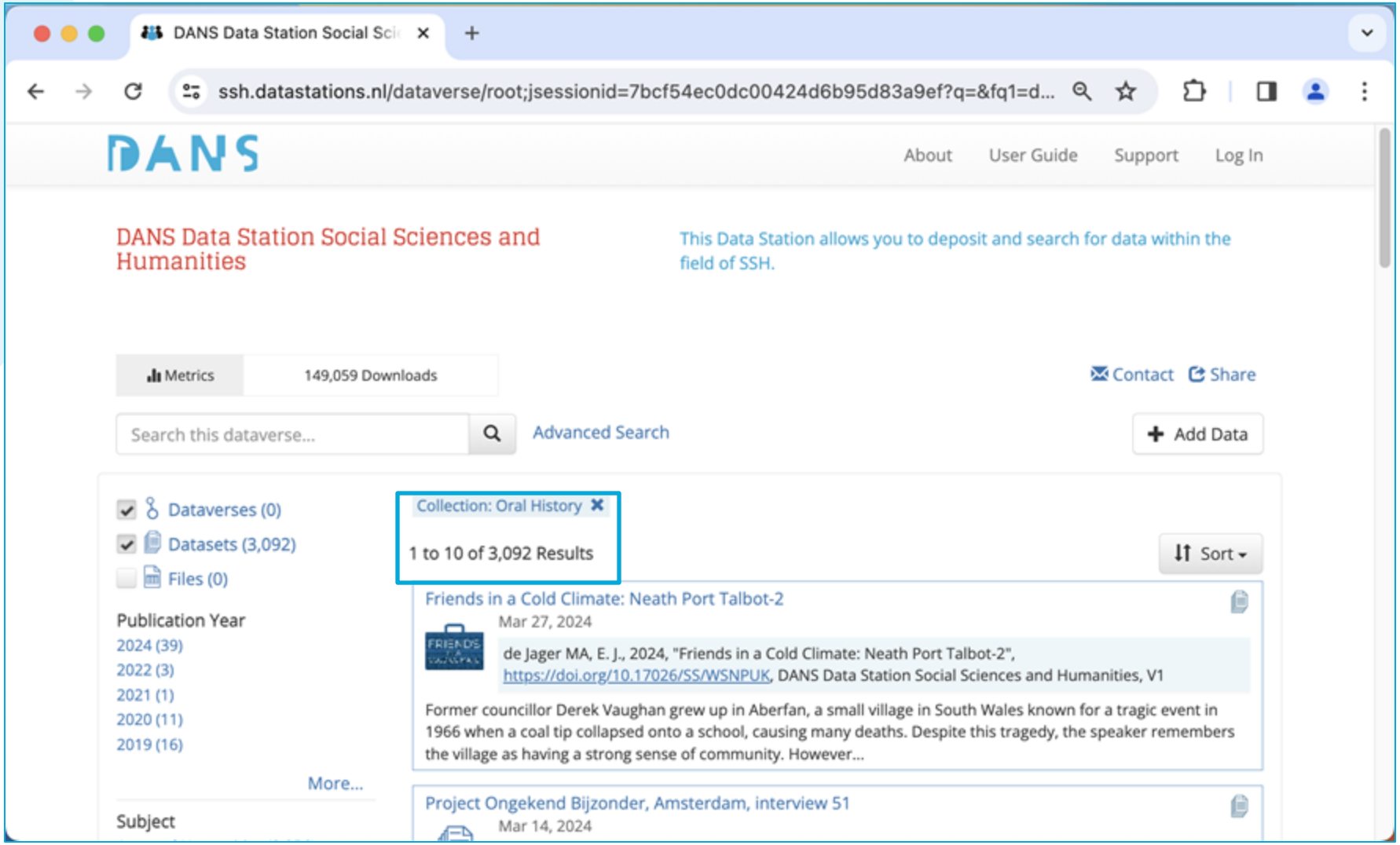


Anyone can **find** data for **reuse**

We **archive and publish** data with involvement from Dutch researchers or focusing on the Netherlands

https://ssh.datastations.nl

Archiving and publishing of qualitative data



Qualitative Data

As open as possible as closed as necessary

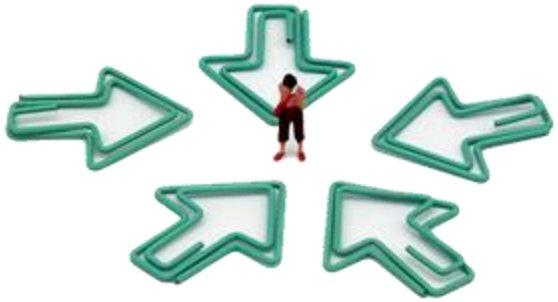
Following the Open Science principles you want to publish Open Access where possible

As open as possible as closed as necessary

Following the Open Science principles you want to publish Open Access where possible

Why would that not be possible?

- Personal data
- Sensitive information



As open as possible as closed as necessary

Following the Open Science principles you want to publish Open Access where possible

→ Share **what you can** share

→ Many repositories offer options publish **under restricted access**!

The challenges of qualitative data

Qualitative data refers to information that is not gathered in numerical form, but rather describes qualities or characteristics.

Image CC-BY-SA https://dmeg.cessda.eu/

The challenges of qualitative data

(Personal) data which is hard to de-identify!

- You lose contextual information
- Qualitative aspects are what you want to study

A lot of existing guidance applies to *quantitative* data

→ More guidance needed!

Qualitative data refers to information that is not gathered in numerical form, but rather describes qualities or characteristics.

Image CC-BY-SA https://dmeg.cessda.eu/

Making qualitative data reusable - Report

https://doi.org/10.5281/zenodo.8160880

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Many aspects to consider…

Planning the study

- ★ Gathering informed consent

Processing data

- ★ Using Open Tools to process your data
- ★ De-identification of your data

Publishing data

- ★ Selecting a suitable repository
- ★ Chosing between open vs. restricted access
- ★ Publishing restricted access

Many aspects to consider…

Planning the study

★ Gathering informed consent

Processing data

★ Using Open Tools to process your data
★ De-identification of your data

Publishing data

★ Selecting a suitable repository
★ Chosing between open vs. restricted access
★ Publishing restricted access

The other aspects are described in the guidebook (available at https://doi.org/10.5281/zenodo.8160880) and there are extra slides at the end of this presentation (available at https://doi.org/10.5281/zenodo.14140251)

Informed consent

Informed consent

- Not only for participation in the study, but also for the archiving and sharing of the data
- Consider in advance what data sharing you want to facilitate!
  - Purposes or role of reuser
  - Which parts of the data for which use (granular consent)
  - Can the data be de-identified?

- Personal data → GDPR
  - Art. 7 > Conditions for Informed Consent
  - Ch. 3 > Rights of the data subject

You can obtain consent for sharing non-anonymised data, but there might still be ethical reasons to not share it openly.

Example templates

Written informed consent process (including online surveys)

Description	File download
Any relevant advertising or recruiting material (poster, email text, social media advert)	> Template poster advert
Template information sheet	> Par info tem
Template written consent form	> Written consent form template
Template information sheet for online research For online tasks only, where there is no face-to-face contact with human participants	> Template information sheet for online research

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Examples and templates

Note that all examples below assume that they are preceded by sufficiently specified information.

Template in Qualtrics | Examples CESSDA

Example sentences

Data Management Expert Guide

Consent needs to be freely given, informed, unambiguous, specific and by a clear affirmative action that signifies agreement to the processing of personal data.

⊖ Click to see examples of consent forms

⊕ UK Data Archive

⊕ MRC Cognition and Brain Sciences Unit - University of Cambridge

⊕ FORS (Swiss Centre o

TEMPLATE 1: Participant Information/Opening Statement

Key points to include	Suggested text
1. Level (eg: Masters, PhD, research) purpose, potential outcomes and implications of the study 2. The role of TU Delft and any third parties including funding body 3. Who participants are (eg: children, experts, students in a dependent role to the researcher) 4. What exactly what they are being asked to do 5. What if any Personal Data (Personally Identifiable Information and/or Personally Identifiable Research Data) will be collected, and how it will be used, published and include clarity on: you collect will be used during the research personal information, maintaining confidentiality (pseudo/anonymising) data ccess to data, data archiving and reuse ta publication and dissemination, and and the retention period for research data or criteria used to nal or reputational risks might arise from participation either y, and what steps will be used to mitigate these risks efuse to answer/withdraw from the study at any time 8. The right (or otherwise) of participants to request access to and rectify or erase personal data 9. Any remuneration for time/compensation for travel 10. Contact details of the Responsible Researcher and procedure for making complaints. **Note: the TU/D Human Research Ethics Committee should not be included as a contact and does not deal with participant complaints.**	You are being invited to participate in a research study titled [*Name of your research*]. This study is being done by [*Name of Researcher(s)*] from the TU Delft [*include also any collaborating partners including internship provider and/or funding body*]. The purpose of this research study is [*provide participants with a short statement about the research*], and will take you approximately [*XX*] minutes to complete. The data will be used for [*provide list of intended uses, including publication, application and teaching*]. We will be asking you to [*provide summary of what kinds of questions or tasks participants will be faced with*]. As with any online activity the risk of a breach is always possible. To the best of our ability your answers in this study will remain confidential. We will minimize any risks by [*be clear on whether the survey is completely anonymous, and/or whether IP addresses or other Personal Data will be collected. If so describe how you will safely store data, how confidentiality will be secured and how it will be anonymised*]. [*mention Open data specifically if applicable*] Your participation in this study is entirely voluntary and you can withdraw at any time. You are free to omit any questions. [*include also clarification on whether data can be removed within a given timescale. This will not be possible where surveys are completely anonymous*] [*Provide contact details for corresponding and Responsible Researcher*] [*If participants are agreeing to this Opening Statement by clicking through to an (anonymous) online survey, this should also be clear in the Opening Statement.*]

TU Delft Delft University of Technology

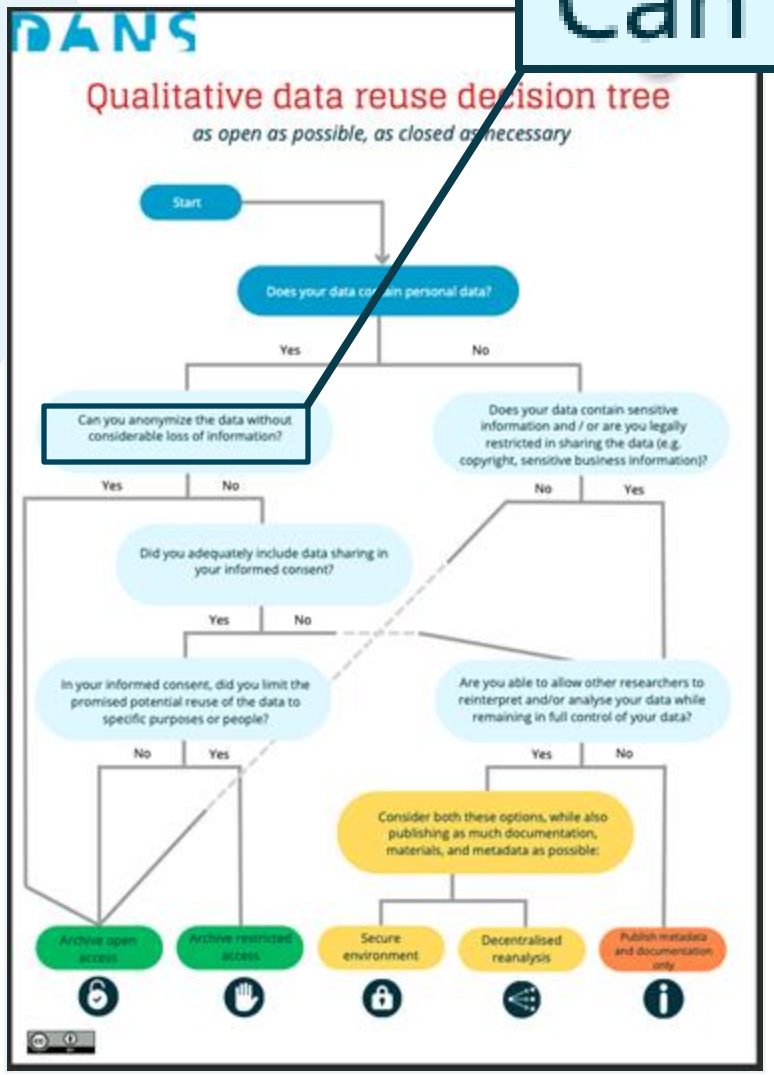
Utrecht University

University of Oxford | CESSDA DMEG | Utrecht University (2) | TU Delft

Deciding on the right data access category

Making qualitative data reusable - Decision Tree

Can you anonymize the data



Qualitative data reuse decision tree

as open as possible, as closed as necessary

Archive open access

Your data is not sensitive, and if your dataset contains personal data you have informed consent that allows the open sharing of data, or the personal data can be anonymised adequately.

Consider to archive your data open access in trustworthy digital repository like the DANS Data Station SSH. CC0 license is recommended. Include metadata and documentation to improve the reusability.

Archive restricted access

Your data contains personal information that you are not allowed to share open access, but your informed consent allows data sharing with specific conditions.

Consider to archive your data in trustworthy digital repository that allows restricted access, like the DANS Data Station SSH. Include metadata and documentation to improve the reusability. Establish a Data Access Protocol to accept or reject requests.

Secure environment

Your data is sensitive and/or contains personal data and you don't have informed consent to share it. Consider using secure environments where the data remains under your control.

Publish project documentation and metadata about your data to make it findable.

Decentralised reanalysis

Your data is sensitive and/or contains personal data and you don't have informed consent to share it. Consider offering your data for decentralised reanalysis, where you reanalyse your data based on analysis protocols from other researchers.

Publish project documentation and metadata about your data to make it findable.

Publish metadata only

Your data is sensitive and/or contains personal data and you don't have informed consent to share it. Secure environments or decentralised reanalysis are not possible for you.

Publish project documentation and metadata about your project to make it findable.

This decision tree poster is part of the Guidebook: "Making qualitative data reusable"
https://doi.org/10.5281/zenodo.7777519

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Data Flow Icon: SBTS from Noun Project,(CC BY3.0)
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Pseudonymisation versus Anonymisation

- In both cases, personal data is replaced

For a comprehensive explanaition see: https://utrechtuniversity.github.io/dataprivacyhandbook/pseudonymisation-anonymisation.html

Decision tree

Personal data

Anonymisation

Consent

Sensitive data (non-personal data)

Personal data

Anonymisation

Consent

Sensitive data (non-personal data)

Personal data

Anonymisation

Consent

Sensitive data (non-personal data)

Archive open access

CC ZERO

CC BY

https://doi.org/10.5281/zenodo.8160890

Decision tree

Personal data

Anonymisation

Consent

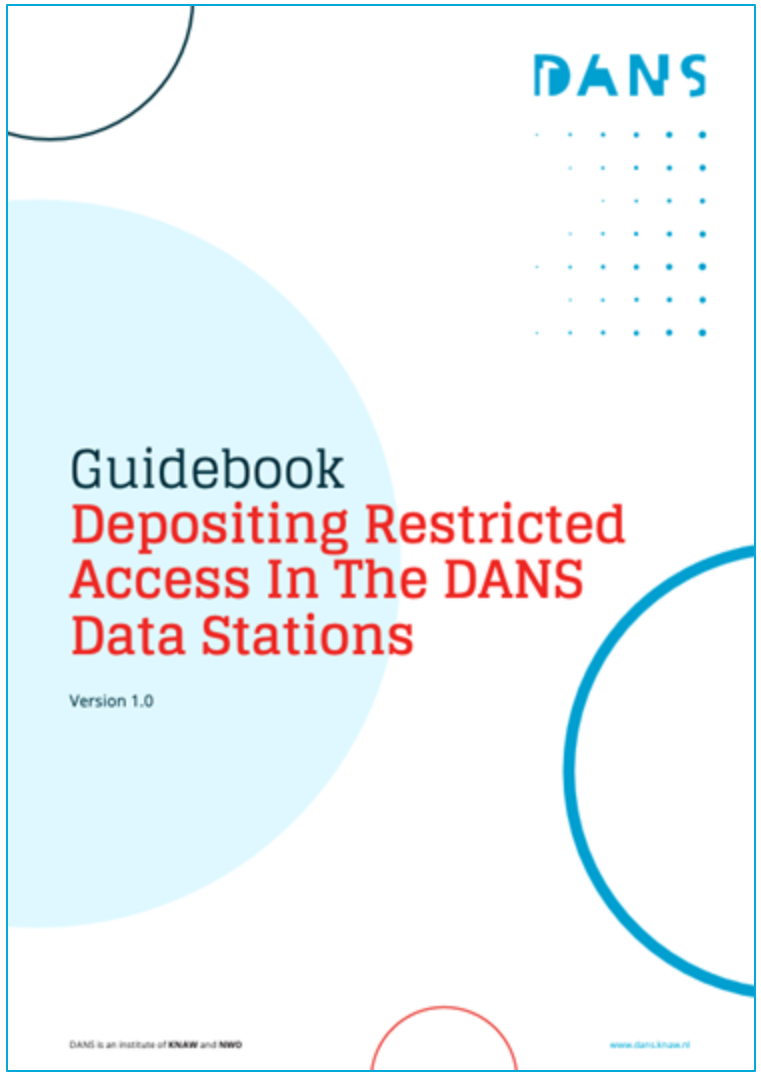
Sensitive data (non-personal data)

Archive
restricted access

https://doi.org/10.5281/zenodo.8160890

Decision tree

Available guidebook
Braukmann, R., Verburg, M., & Mahabier, W. (2024). Guidebook - Depositing Restricted Access In The DANS Data Stations (1.0). Zenodo. https://doi.org/10.5281/zenodo.10887484



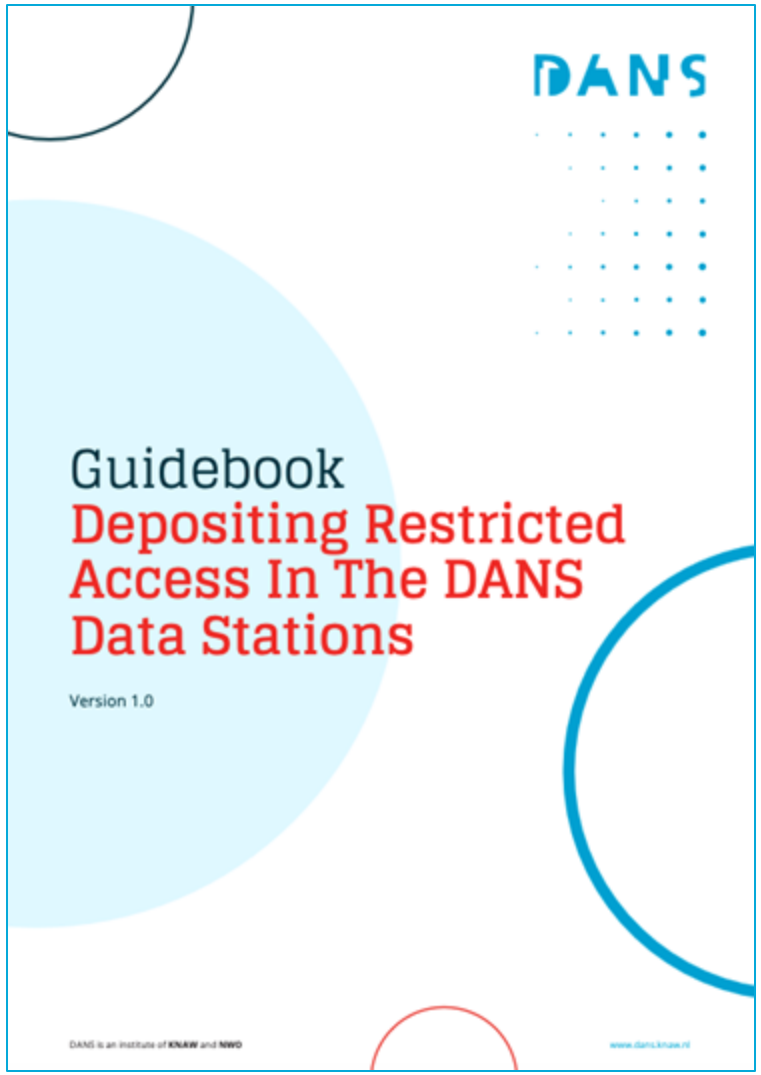
Archive restricted access

https://doi.org/10.5281/zenodo.8160890

Guidebook on Restricted Access

- Focus on depositing in the DANS Data Station SSH
  - Access categories and licences we offer
  - Metadata fields around access information
  - This may not always apply to other repositories!

- Does include general guidance
  - Tips on selecting the right access for your dataset
  - Tips on what information to document
  - Template for a Data Access Protocol



https://doi.org/10.5281/zenodo.10887484

Prepare a Data Access Protocol

- DAP template takes you through all the questions to consider when restricting access
  - Do you allow
    - use beyond research?
    - commercial use?
    - educational use?
    - use outside of Europe?
  - Do you require a motivation for reuse? How is it evaluated?
  - Who is responsible to evaluate requests? How is this organized now and in the future?

Download DAP template:
https://doi.org/10.5281/zenodo.10887571

- Preparing this well makes it easier for you to manage access
- Researchers who want to reuse data can better evaluate if they are allowed to

Questions?

Contributors

Making Qualitative Data Reusable

- Maaike Verburg - DANS
- Widia Mahabier - DANS
- Jetze Touber – DANS
- Samantha Willemsen – DANS
- Hans Berends - VU
- Eric Haynes – VU
- Kacana Khadjavi Pour – VU
- Fleur Deken - VU

Questions?

Thank you for your attention

Visit our website **www.dans.knaw.nl**

And follow us online

Mastodon @DANS_knaw_nwo

LinkedIn @DANS

X @DANS_knaw_nwo

Additional Slides

Planning the study

- ★ Gathering informed consent

Processing data

- ★ Using Open Tools to process your data
- ★ De-identification of your data

Publishing data

- ★ Selecting a suitable repository
- ★ Choosing between open vs. restricted access
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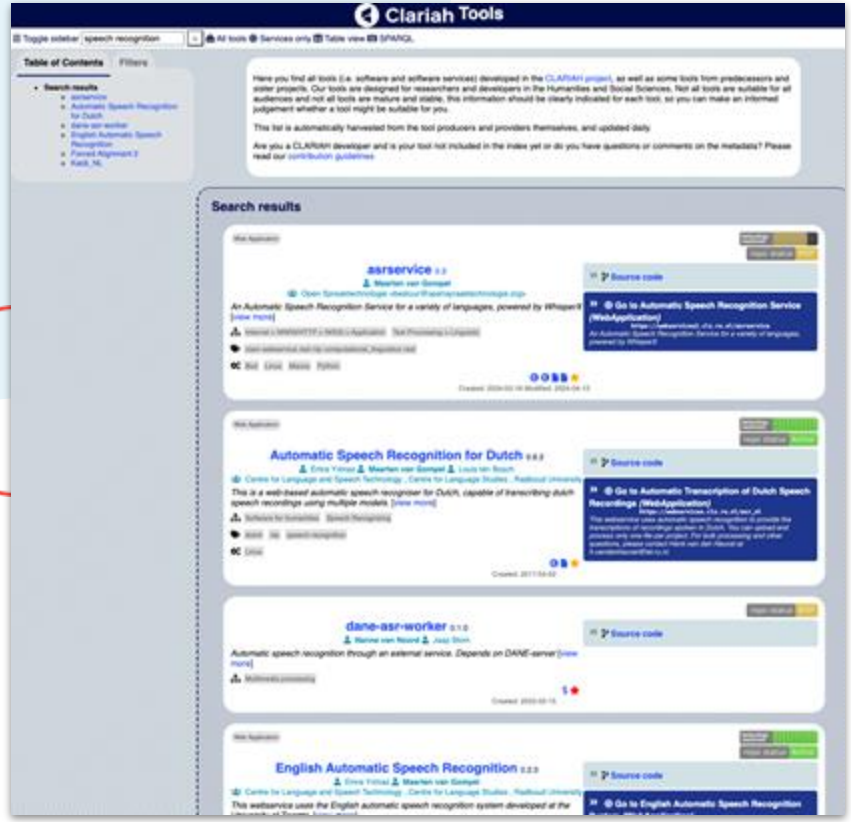
Open Tools for processing

Open tools for transcription

CLARIAH tools - Catalogue of tools and services for different purposes. ASR (Automatic Speech Recognition) tools in different languages and based on different scripts/softwares



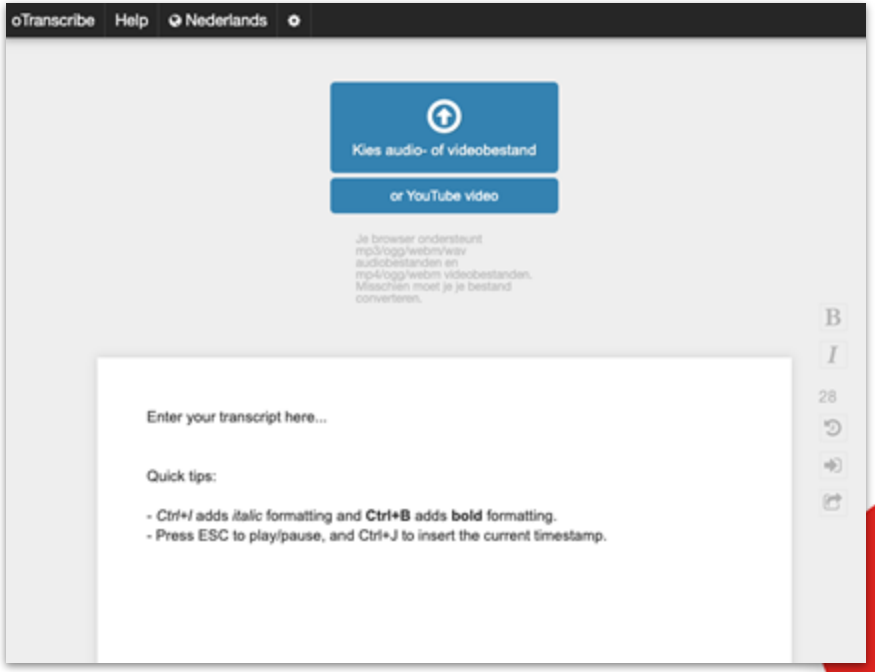
Python packages - Open and flexible, but require some programming knowledge

Picking a Python Speech Recognition Package

A handful of packages for speech recognition exist on PyPI. A few of them include:

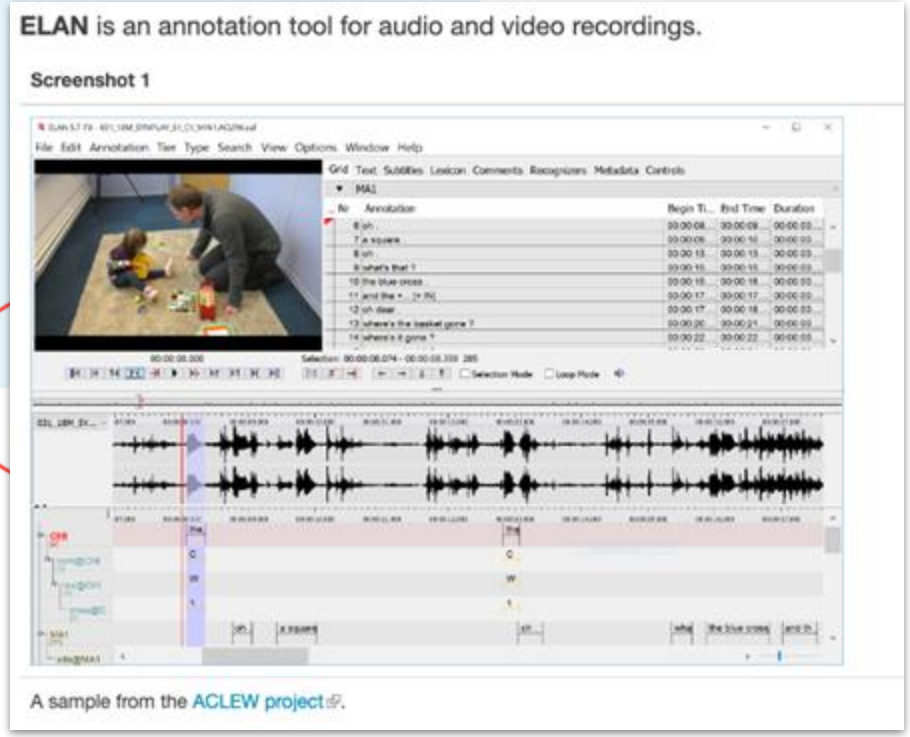
- apiai
- assemblyai
- google-cloud-speech
- pocketsphinx
- SpeechRecognition
- watson-developer-cloud
- wit

oTranscribe - Supporting tool for transcribing manually. Files stay local so no data risk. Multiple languages for the interface.



Open tools for annotation

ELAN - free annotation tool for audio and video recordings developed by The Language Archive from the Max Planck Institute for Psycholinguistics. Open GPL3 license.



Taguette - Free and open-source tool to support manual processing of text. Work locally for data security.

Hypothes.is or **ATI** - for annotating literature / publications in pdf format. Can be collaborative.

Open tools for anonymisation/pseudonymisation

anonymoUUs - Python package, MIT license. Runs through entire file tree to replace information with pseudo-IDs.

UKDA anonymisation helper - MS Word add-on. Does not change data, but highlights identifiable information to be considered manually.

Textwash and **FAMTAFOS** - Automatic identification and replacement of identifiable data. GPL3 license.

Utrecht University

anonymoUUs

anonymoUUs is a Python package for replacing identifiable strings in multiple files and folders at once. It can be used to pseudonymise data files and therefore contributes to protecting personal data.

UK Data Service

Tools for anonymisation

Our text anonymisation helper tool can help you find disclosive information to remove or pseudonymise in qualitative data files. The tool does not anonymise or make changes to data, but uses MS Word macros to find and highlight numbers and words starting with capital letters in text. Numbers and capitalised words are often disclosive, e.g. as names, companies, birth dates, addresses, educational institutions and countries.

Textwash

Textwash is an automated text anonymisation tool written in Python. The tool can be used to anonymise unstructured text data. To achieve this, Textwash identifies and extracts personally-identifiable information (e.g., names, dates) from text and replaces the identified entities with a generic identifier (e.g., Jane Doe is replaced with PERSON_FIRSTNAME_1 PERSON_LASTNAME_1).

De-identification

Deidentification of data

- Removing or replacing identifiable information from the data
- To increase possibilities of data sharing and reuse
- **Anonymous** - data cannot be re-identified (GDRP does not apply anymore)
- **Pseudonymised** - re-identification of the data remains possible (e.g. through a key)
- Planning for data reuse
  - data minimisation
  - creating de-identification pipeline
  - granular consent (choice for data subjects what to accept)

→ Guides, tips & tricks: **FSD** | **UKDA** | **CESSDA** | **Erasmus University**

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So far available in English and Dutch only

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Selecting a repository

Selecting a repository

- Archive data so it is securely stored for the long term
- Publish (meta)data so the dataset can be found and reused

→ Selecting a Trustworthy Digital Repository

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- Archive data so it is securely stored for the long term
- Publish (meta)data so the dataset can be found and reused

→ Selecting a Trustworthy Digital Repository

- Your institute may have regulations or recommendations

CORE TRUST SEAL

Preparing your dataset

- Prepare your dataset according to the repository guidelines
  - Metadata and Documentation
  - Preferred file formats

- Available guidance (for qualitative data)
  - CESSDA - Data Management Expert Guide - Documentation and Metadata
  - DANS - Deposit manual
  - DANS - Preferred file formats
  - FSD - Processing qualitative data files

→ **Decide** whether your data can be published **open** or under **restricted access**